

Career

“SEMICON MANUFACTURING Has The Potential To Generate A Huge Market For Jobs In Both Skilled And Unskilled Fields”

As part of being Atmanirbhar Bharat, India is working towards setting up semiconductor manufacturing facilities and is receiving proposals from big players like Vedanta, Foxconn, IGSS, and ISMC. Chetna Mohan, President, El Camino Technologies Pvt Ltd, spoke to Abhinaya Kuzhanthaivel of EFY about the career opportunities that this opens up for graduates in electronics and electricals, physics, applied physics, material sciences, chemical, and mechanical disciplines, and the skillsets they need to build



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How do you see the semicon manufacturing space evolving in India?

I would categorise semicon manufacturing into different sectors for ease of understanding, and each one is an individual unit on its own. These are:

- Design houses
- Wafer fabs (including raw wafer manufacturing and crystal growth)
- Assembly lines that include assembly, testing, marking, and packaging (ATMP)
- Failure analysis and reliability analysis (FA and RA) labs

India is a hub of the semiconductor design industry having well-established R&D and design centres for VLSI design, board design, and embedded software companies. Major players across the world have their R&D and design centres in India. However, raw wafer manufacturing and crystal growth are yet to be established. For fabs, the investment is huge and return on investment (ROI) takes a long time, whereas packaging lines are more flexible and easier to set up as compared to fabs. To be self-reliant, India will need wafer manufacturing, assembly, and testing facilities.

What do you think about the type of jobs and skillsets that will be in demand if semicon manufacturing picks up?

Semiconductor manufacturing needs a wide range of skillsets when we see at a macro level. It has the potential to generate a huge market for jobs in both skilled and unskilled fields. It could target youth from rural areas to scientists, designers, researchers, process developers, quality control, utilities, and facilities experts. The industry needs different levels of workforce

for production, including shop-floor assistants, helpers, warehouse management, security, etc.

Apart from the production related skills, there are other add-ons and supporting clusters of industries which grow when any other production unit grows, such as suppliers of raw materials, chemicals, high purity gases, gas handling systems, water handling equipment, electricity, cleanrooms, related furniture, pollution control, disposal systems, chemical treatment plants, environment controlled facilities, certification bodies, which will require both Indian and overseas products and expertise.

Do we have courses or curriculum that gives on-hand training before getting to the job?

As I understand, we do not find a defined curriculum, particularly on microelectronics, including semiconductor technology in detail, at undergraduate levels in most universities. Premier institutes, such as IITs and IISc, have well-established labs where students have exposure. Some of the labs at premier institutes include labs for basic electronics, microelectromechanical (MEMS) systems, microelectronics, VLSI engineering, MEMS and microsystems, microwave and millimeter-wave circuits, nanofabrication, nanophotonics, nanoscale optoelectronics, semiconductor devices and circuits, which gives the students preliminary understanding of the subjects.

We do see a few chapters on VLSI designing and semiconductor devices, but complete process and know-how of semiconductor manufacturing and fabrication are not available at graduation levels. As I have seen in my experience in this area, most of the training is on-the-job. Since this is a high-tech industry, it takes six to nine months before an